

SECTION F

Technical Specification

1. DIGITAL VIDEO COLPOSCOPE

All items must be from one manufacturer for smooth compatibility and working of the system. Colposcope should be USFDAEU-CE/BIS Certified from same manufacturer

Full HD Colposcope

- Should be Full HD Technology with CMOS Sensor
- Should be 1080p Resolution Sony Sensor (1920x1080
- Should be Aspect Ratio16:9
- Should be 21.2 Mega Pixels
- Should be No need of Dongle & Capture Card
- Should be Multiple Grades of green Filter
- Should be Recommended Lamp life 50000 hours
- Should be Highest quality 1-52x Magnification
- Should be Progressive Scanning Lines
- Should be Latest Digital Matrix Processor
- Should be Fine Advance & fast Auto/Manualfocus- Fine Technology
- Should be E Flips & Mirror image Facility
- Should be Display of genuine tissue color and tone
- Should be Corner to Corner Uniform, Brightness
- Should be Real time display of magnification
- Should be Acetic acid Test Timer
- Should be Electronic Green Filter with no light loss
- Should be Black Morphological Filter.
- Should be Minimum illumination 0.1 Lux.
- Should be Remote Control Facility
- Multi-function Image processing software.
- Should be Integrated Management of View Examination, image capture &Freeze, Recording, Observation, Processing.
- Should be Can be integrated to LAN and HIS
- Should be Reference pictures library to help user
- Should be Statistics mode with automatic flow chart facility is provided.
- Should be Printing of Multi Format photo report with Hammond graph Report
- Should be Powerful Comparison to compare
- Should be Colposcopy REID Evaluation
- Should be Report can be e-mailed as well
- Computer with printer, ups, trolley
- Should Have 21"color TFT flat panel Monitor
- Should Have Microsoft Windows 10
- Should Have 500GB HardDisk
- Should Have IntelCorei7 Processor or better

- Should Have 8 GB Ram
- Inkjet Coloured Printer
- Trolley (Indian Make).
- One year warranty

2. DOUBLE PUNCTURE LAPROSCOPE SET

Telescope	
5-7mm 30 Degree HD Telescope Autoclavable Fiber optic light transmission incorporated should be compatible with the commonly available fibre optic light cables (necessary adaptors should be provided) fit to purpose.	1
Trocar r& Cannula for telescopes	2
Resusable metal trocar and Cannula size +0.5 to 1mm more than the telescope diameter; should have an automatic valve and stopcock for insufflations. Trocar should have pyramid tip 10-11.cm.	
3 (a) Ring Applicator	2
Ring applicator double puncture approach, for use the Trocar size 5 to 7mm capable of loading two silastic rings.	
3(b) Trocar and Cannula for Ring Applicator	2
Cannula size + 1 mm more than the ring applicator. Trocar should have pyramidal dip. Antifogging solution (100 ml.) to be provided with both. The length of the Trocar should be 160-170 mm and the length of the cannula should be 100-110mm.	
Cone	5
Suitable cones for loading rings to the above applicator	
Slide/guide	10
Suitable slides/guide for loading rings to the above applicator	
Veress Needle	3
Veress Needle with spring loaded blunt stylet with luer lock and length of 10 cm	
Veress Needle	
Veress Needle with spring loaded blunt stylet with luer lock and length of 15 cm	3
Essential Spares	
i) Spare Washers	12
Spare washers for trocar and cannula and automatic valve	
ii) Kits for cleaning	
i) Trocar Brush	2
ii) Cannula Brush	2
iii) Cleaning Oil 100 ml	2
iii) Manuals	

Operational manual Spare parts list (with part nos.)	1
Specification for CO2 Insufflator and Cold Light Source for Double Puncture Laparoscope	
Carbon Dioxide Insufflators	1
Electronic CO2 insufflator	
Should have a continuous adjustable flow rate from 0 to 20 ltr in minor increments and a pressure range continuous adjustable between 0-30 mm Hg. Set Pressure and Set flow rate & Actual Pressure & Actual Flow rate should be displayed on the front panel- Provided with Silicon autoclavable tubing with luer attachment. Instrument should work on a supply of 220-240 V, with a frequency of 50 Hz single phase Should have a continuous adjustable flow rate from,0 to 20ltr in minor increments and a pressure range continuous adjustable between 0-30 mm Hg. Set Pressure and Set flow rate & Actual Pressure & Actual Flow rate should be displayed on the front panel - Provided with Silicon autoclavable tubing with luer attachment. Instrument should work on a supply of 220-240 V, with a frequency of 50 Hz single phase	
Should be provided with audio or Visual alarms to identify and alert the Low Pressure and occlusion in the patient tubing, during the surgery	
The model should have the following features –	
1. Automatic release of excess gas in case of excess pressure in the abdomen	
2. Available gas in the cylinder in progressive bar graph should be displayed or give audiovisual alarm. This would ensure the Doctors would be aware the amount of gas available	
3.The set and actual flow rate and pressure should be displayed	
4.The total gas consumed should be displayed	
5. The unit should be of at most safety and standard and should GE & IEC 60601 Certified	
High Pressure Hose	1
Suitable High Pressure Hose, easy availability of the spare in India.	
Wrench Kit	1
Suitable for connecting the insufflator with CO2 cylinder	
Carbon Dioxide Bottle	1
B type cylinder 7 Kg capacity carbon-di-oxide cylinder to be supplied with all safety certification governing the local Indian certifications	2
Cold Light Source	
LED minimum of 175 watts or more light source with long bulb life, CE & IEC 601-1 certified.	1
Fibre Optic Light Cable	1
4.5-6.5 mm in diameter, 180-200 cm in length compatible with the cold light source and the commonly available telescopes	

UPS	1
UPS-1KVA with one hour backup time	
Formalin Chamber (for Sterilization of Laparoscopes)	2
Tray for sterilization of Laparoscopes	1
One year warranty.	

3. **HYSTEROSCOPE SET**

4.0mm 30° High Definition Hysteroscope -1No

I. Hysteroscope Detailed specification:

- Should have Proprietary optical design and anti-reflective lens system to deliver superb image resolution
- Should have the sapphire glass to hard enough to protect the sophisticated lens system from accidental cutter and burs scrapes and to give best image.
- Should have the Proprietary light cone and fibre bending technologies to deliver glare free illumination across the entire field of view.
- Should have the Superior Field view in HD Quality
- Should have the Improved depth of focus for viewing the at the depth of the site without any problem.
- Should have the Laser welded joints for the better Autoclaving
- Should be Used with Resectoscope Sheaths.
- Should have CE/USFDA/BIS certified.

II Sheath and instruments

- Inner Diagnostic sheath for 4.0 mm Hysteroscope 1No
- Outer Diagnostic sheath for 4.0 mm Hysteroscope 1No
- Inner Operative sheath for 4.0 mm Hysteroscope 1No
- Outer Operative sheath for 4.0 mm Hysteroscope 1No
- Operative and Diagnostic Obturator 1each
- Semi Flexible Scissors Ø1.6 mm, 5 Fr. WL 340mm, single action 1No
- Semi Flexible Grasping Forceps Ø1.6mm, 5 Fr., WL 340mm, double action 1No

III Resectoscope Set

- 24/26Fr Continuous Flow Resectoscope Sheath Set (Inner/Outer) 1No
 - Resectoscope Standard Obturator 1No
 - Working Element-Active (Bipolar) 1No
 - Disposable Port Closing Seals 10No
 - 24Fr. cutting loop bipolar electrode 6Nos
 - 24Fr.Collings Knife bipolar Electrode 6Nos
 - Bipolar Electrode and Cable 1No
 - High Frequency Cord 1No

Advanced High Definition Camera System with 45Ltr High Flow Insufflator.

S.No.	Description	Qty
1.	Telescopes for both Operative & Diagnostic laparoscop	
(i)	HD Telescopes 10mm 0 Deg	1
	High definition wide Angle Full Screen, Forward-Oblique and lateral scope, Optimal center-to-edge resolution for enhanced picture quality, Fiber optic light transmission incorporated, Standard ocular window for coupling the camera head, Scratch resistance sapphire quoted tip lens, Shguld be ready for ICG and special imaging mode for visualization of real time imaging of pertusion, blood flow, and biliary mapping and standard white light	
	Angle of view: 0 Diameter 10 mm, Length 28-30cms	
(ii)	HD Telescopes 10mm 30 Deg	1
	High definition wide Angle Full Screen, Forward-Oblique and lateral scope, Optimal center-to-edge resolution for enhanced picture quality, Fiber optic light transmission incorporated, Standard ocular window for coupling the camera head, Scratch resistance sapphire quoted tip lens, Should be ready for ICG and special imaging mode for visualization of real time imaging of perfusion, blood flow, and biliary mapping and standard white light	
	Angle of view: 30° Diameter 10 mm, Length 28-30cms	
2	4K Ultra-HD 3 CCD / 3 Chip CMOS Digital Camera/ Processor and Camera Head	1
i	It should have pure digital signal with high definition video of 3840x2160p (min) native resolution and progressive scan technology both on camera head and console, Should have controls on camera console and camera head.	
ii	It should be compatible with Aspect ratio of 16:9	
iii	Should have special imaging mode like visualization of real time imaging of perfusion, blood flow, and biliary mapping and should also have ICG mode	
iv	The system should Digital zoom with ICG complaint	
v	Camera Head should have 4 buttons for features like zoom, white balance, ICG fluorescent viewing, for special imaging mode	
vi	camera head buttons for toggling between special mode for visualization of real time imaging of perfusion, blood flow, and biliary mapping and standard white light	
vii	Camera head should have feature to decreases the saturation of color in the image to the level preferable by the surgeon	
viii	Integrated Gain/Shutter/Enhancement with automatic brightness control	
ix	Should have these output ports: 2 X HDMI/ 3G-SDI [Signal Format: 3840x2160, 1080p]	
x	The system should automatically optimize all settings. The system should be ready to- use as soon as it is connected to the camera control unit.	
xi	The system should be Menu driven, thus allowing the surgeon to program the camera head functions as per the surgical needs & requirement.	

xii	The Camera & Coupler which can be used with any make of standard laparoscopes, cystoscopes and resectoscopes	
3	LED light source with 300 watts or more	1
i	Enables users to visualize standard white light wavelengths, Infrared light wavelengths, and Endoscopic Near Infrared wavelengths using ICG technology	
ii	Should have adjustable light intensity in both white light mode and in special imaging mode for visualization of real time imaging of perfusion, blood flow, and biliary mapping and standard white light	
iii	Should have integrated infrared lighted fibres for ureteral detection, if any additional accessories for same is required then that should be provided.	
iv	Increased patient safety & added protection in OR with intuitive simple user interface with LED/LCD touch screen	
v	Standby Mode	
vi	Universal Jaw Assembly to adapt any make of Fiber Optic Cable	
vii	Lamp life of 5000 hrs, as standard or with spare bulbs	
viii	Should be controlled through HD camera also	
4	Fiber Optic Light Cable	1
i	Size should be diameter 4.5mm or more, length 160 cm or more	
ii	Should be ready for ICG and special imaging mode for visualization of real time imaging of perfusion, blood flow, and biliary mapping and standard white light	
5	4k Medical Grade Monitor	1
i	4k 32" or more Flat Panel Monitor	
ii	PAL system compatible	
iii	HDMI, DVI Inputs, 3G-SDI inputs	
vi	Compact & Lightweight design	
V	Native Resolution 3840x2160p or above.	
Vi	The monitor should support Direct Fiber input for long display transmission	
6	IRIS Ureteral Stenting Kit	1
7	Insufflators	1
	High flow of 45ltr or above with LED/LCD display	
	Microprocessor controlled & Software driven for upgradeability	
	Soft approach pressure control for safe recovery of abdominal pressure	
	Should have Neonatal mode & visual and audible alarms with min 0.1 L flow rate internal leakage detection capability	
	Integrated Gas heating, having internal venting system for safety	
	Should include heated tubing, hose & yoke	
8	4K Image and Video Management System with 15.6 " Touch Screen Monitor	1
	The resolution should be 3840x2160p (min) native resolution , internal capacity of 1TB minimum, should be controlled through camera head for still and videos in full 4K resolution, Should have separate 15.6" Medical Grade Monitor with touchscreen.	
9	Trolley with Flexible Arm to mount the monitor (Indiani	1

	Trolley should be capable to mount above-mentioned system	
	Should have space to keep CO2 cylinder	
	Should have movable arm to mount monitor	
	All the items except the Indian Make should be USFDA /European CE certified/BIS certificate should be submitted with tender documents.	
	Warranty :One year	

4. PROCTOSCOPE

- Made of non-rusted high grade stainless.
- Can be autoclave, Re Sterile, Reusable.
- Smooth Tip for easy insertion in anal.
- Standard should be USFDA/CE/BIS and CDSCO and ISO

5. SINGLE PUNCTURE LAPROSCOPE SET

DESCRIPTION	Qty
Telescope	
10mm 0 Degree HD Telescope, with parallel eyepiece, wide angle 270mm length, with 6mm instrument channel Autoclavable Fiber optic light transmission incorporated should be compatible with the commonly available fibre optic light cables (necessary adaptors should be provided) fit to purpose.	1
Trocar & Cannula for telescopes	1
Reusable metal trocar and Cannula size +0.5 to 1mm more than the telescope diameter; should have an automatic valve and stopcock for insufflations. Trocar should have pyramid tip 10-11 cm.	
Ring Applicator	1
Ring applicator double puncture approach, for use the Trocar size 5 to 7mm capable of loading two silastic rings.	
Cone	2
Suitable cones for loading rings to the above applicator	
Slide/guide	1
Suitable slides/guide for loading rings to the above applicator	
Veress Needle	1
Veress Needle with spring loaded blunt stylet with luer lock and length of 10 cm	
Veress Needle	
Veress Needle with spring loaded blunt stylet with luer lock and length of 15 cm	1
Essential Spares	
<i>i) Spare Washers</i>	5
Spare washers for trocar and cannula and automatic valve	
<i>ii) Kits for cleaning</i>	
i) Trocar Brush	2
ii) Cannula Brush	2
iii) Cleaning Oil 100 ml	2

<i>Hi) Manuals</i>	
Operational manual Spare parts list (with part nos.)	1
Specification for CO₂ Insufflator and Cold Light Source for Single Puncture Laparoscope	
Carbon Dioxide Insufflators	1
Electronic CO₂ insufflator	
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High Pressure Hose	1
Suitable High-Pressure Hose, easy availability of the spare in India.	
Wrench Kit	1
Suitable for connecting the insufflator with CO ₂ cylinder	
Carbon Dioxide Bottle	1
B type cylinder 7 Kg capacity carbon-di-oxide cylinder to be supplied with all safety certification governing the local Indian certifications	2
Cold Light Source	
LED minimum of 175 watts or more light source with long bulb life, CE & IEC 601-1 certified.	1
Fibre Optic Light Cable	1
4.5-6.5 mm in diameter, 180-200 cm in length compatible with the cold light source and the commonly available telescopes	
UPS	1
UPS - 1 KVA with one hour backup time	
Formalin Chamber (for Sterilization of Laparoscopes)	2
Tray for sterilization of Laparoscopes	1
1 year warranty	

6. VIDEO OPTIC LARYNGOSCOPE WITH 4 REUSABLE BLADES

- Display Size should be 3.0" LCD Monitor or better.
- Screen Resolution 640 x 480 (RGB) or better for video quality.
- Battery working time ≥ 240 min, High-energy lithium battery
- Should come with different size reusable blade (Blade size Mac1, Mac2, Mac3 and Mac4)
- Weight of the product should be not more than 225g.
- Screen movement should be Front and Back: 0° - 130° and better.
- Should come with Anti-Fogging camera and blade.
- Should have USB read and store.
- Field Angle $\geq 60^{\circ}$
- Internal memory of the device should be 8 GB
- Should be Product certification: USFDA/CE/BIS and CDSCO.
- 1 year's warranty

7. INTERMITTENT COMPRESSION DEVICE FOR DVT PROPHYLAXIS

- It should have sequential compression pattern.
- Sequential Air Compression Device with REVERSE & SKIP function used for
 - Lymphatic Drainage
 - Lymphatic Edema
 - Varicose Veins
 - DVT(Deep Vein Thrombosis)
 - CVI (Chronic Venous Insufficiency)
 - Vascular Problems
 - Swollen legs
 - Edema
- Should be supplied with standard 6 Chambers Legs fittings, Waist & Arms fittings.
- Standard should be USFDA/CE/BIS and CDSCO
- 1 year warranty

8. OT LED SHADOWLESS LAMP

- Single dome mobile type Operation Theater light shadow-less
- Light output should be minimum 1,60,000 lux at 1 meter distance

- Diameter of the dome should be less than 650mm
- The dome should have single bulb with standby bulb. The standby bulb can be turned on automatically in case of failure of the primary bulb. Should have indication for bulb failure.
- Minimum light field size should be 200mm
- Power supply to the lamp should be from a constant voltage source/SMPS
- Color temperature should be between 5000 EsK or above.
- Should have short circuit protection for the low voltage power supply
- Should have user friendly lamp replacement
- Should have heat reduction filters
- Should have field focus control through sterile handle. Additional sterile handles two numbers should be supplied
- Five spare bulbs should be supplied
- Minimum spring arm stroke of 500mm
- Minimum four step intensity control
- Should be mounted on castors for easy movement
- Input supply-230Vac, 50Hz.
- Standard should be complying with USFDA/CE/BIS and CDSCO
- 1 year warranty

9. OT LIGHT CEILING DOUBLE DOME

The surgical ceiling LED Light Double Dome Should have the following features

1. Each dome should have approximately 150000 LUX.
2. Should have the Digital control panel in the dome to control the light parameters
3. Should have multicolor LED option to varying the colour temperature
4. The illumination depth should be 1200 mm
5. The focus adjustment should be there to varying the field diameter in each dome
6. Should have the colour temperature of 3500-5000^oK
7. Should have the Colour induction index of 97%
8. The size of the light Field Should be 100-300mm
9. Should have the light adjusting mode range from 1-100%
10. The life span of the LED should not be less the 50000 Hours.
11. Power supply should be 220-230v AC.
12. One year warranty.
13. Standard should be complying with USFDA/CE/BIS and CDSCO

10. OT LIGHT CEILING SINGLE DOME

The surgical ceiling LED Light single Dome Should have the following features

1. Each dome should have approximately 150000 LUX.
2. Should have the Digital control panel in the dome to control the light parameters
3. Should have multicolour LED option to varying the colour temperature
4. The illumination depth should be 1200 mm
5. The focus adjustment should be there to varying the field diameter in each dome
6. Should have the colour temperature of 3500-5000°K
7. Should have the Colour induction index of 97%
8. The size of the light Field Should be 100-300mm
9. Should have the light adjusting mode range from 1-100%
10. The life span of the LED should not be less than 50000 Hours.
11. Power supply should be 220-230v AC.
12. 1 year warranty.
13. Standard should be complying with USFDA/BIS/CE and CDSCO

11. OT TABLE WITH STANDARD ACCESSORIES

The OT table should have the following features.

It should have the longitudinal slide of 30 cm with slim central column and low profile base facilitate maximum C-arm access

It should have pressure relief mattresses having patient protection and X-ray attenuation for protecting the health of surgical team when using a C-arm with the following properties

- Watertight
- Flame-retardant
- Antistatic and Antibacterial
- Disperse Pressure to prevent patient bedsores.
- It should have interchangeable sections to operate seamlessly in reverse orientation provide the maximum X-ray translucent distance to the surgeon.
- It should have the Detachable Split leg section for better patient access and can be removed entirely.
- It should have the kidney elevator (Manual)
- It should have the hand pendent
- It should have battery back up
- It should have 800lb weight bearing capacity
- It should have full body imaging facility without special accessories

- It should have support on heavy base column covered with stainless steel sheet
- Standard should be complying with USFDA/BIS/CE and CDSCO

Technical details

Table Top Slide:	300mm
Minimum Height:	750mm
Maximum Height:	1050mm
Length of the Table Top:	2150mm
Width of the Table Top:	500mm
Lateral Tilt:	+20°/-20°
Trendelenburg:	+25°/-25°
Back Section:	+90°/-40°
Head & Leg Section:	+90°/-90° Manual Operated
Kidney Elevator:	150mm Manual Operated

Accessories for OT table

Anesthetic frame	1No
Paddle side support	1pair
Intravenous Arm board	1pair
A paddle shoulder supports	1pair
Knee Crutches	1pair
One Matters Set	Thickness 50mm

12. PORTABLE LED STANDING LIGHT

- Mobile OT Light based on innovative LED technology.
- With 4 casters for smooth movement of mobile light.
- OT Light should have lamp head made of aluminium
- Should have anti-bacterial grip handles on dome for easy rotation and hygiene.
- Should have grip handle on column for easy movement of mobile light
- High fail safety through optical light system
- Single non winged round shape lamp head for easy maintenance and better sealing against dust and fumigation
- Should have Cold, high intensity light through multi lens matrix technology.
- Should have intensity controller to offer different level of light intensity
- Should be strobe free light

- Should have wireless remote for controlling all parameters like field size adjustment, intensity control, on/off, etc.
- Should have override control system for intensity control, on/off, in case of failure of main electronic control panel, to avoid downtime of equipment.
- The unit should be CE /BIS /USFDA and CDSCO certified.
- Intensity (Lux) : 70000 or more
- Color temperature (K): 4500
- Main Power Supply : 220
- Life of LED : 50000 hrs.
- 1 year warranty

13. MOVABLE SHADOW LESS SPOTLIGHTS WITH ADJUSTABLE ARMS

- Economical, Compact, Ultra-high brightness focused flexible light.
- On /off switch integrated with handle.
- Articulated arm / extended handle facilitate position of light.
- IR-free technology from LED light source reduces heat sensation.
- Standard should be USFDA/CE/BIS and CDSCO and ISO
- LED service life 40 times longer than Halogen bulb.
- Low energy consumption: 1/3 less than conventional Halogen system.
- Multi-functional control systems with various installation range such as Floor / wall/ surgical table /desk / etc.
- Easy maintenance, hygienic and cleaning requirements via spray disinfection.
- Suitable for GP practices, E.N.T. Ophthalmology, Gynecology, small theatre, minor operations and other surgery use.

Technical Specifications:-

- Light Intensity : > 70,000 Lux @ D=30 cm / > 40,000 Lux @ D=50cm or better.
- Control System: step less brightness dimmer.
- Flexible arm: 76cm length flexible plastic sleeve goose neck.
- Lamp head Diameter = 75mm, Length = 110mm
- Diameter of Light Spot 80-150mm @ D=30-50 cm or better size.
- Light source 1 x LED bulb.
- LED life span > 50,000 hours or better life.

- Color Temperature 4500K – 5500K or better.
- Input Connected Load 100-260V 50/60MHz
- Power consumption $\pm 5W$.
- Mounting Options Mobile
- 1 year warranty

14. CPAP MACHINE

- Should have Manual and Auto CPAP mode with advanced sensor technology for tracking respiratory status of the patient.
- It should be suited for therapy of OSA / Obesity hypoventilation therapy.
- Should have inspiration trigger for auto start.
- Should have auto stop option when put-off mask.
- Should have alarms for power off, disconnection.
- Should have facility for sensitivity setting.
- Should have leakage compensation facility.
- Should have integrated DC Powered humidifier or equivalent with anti- counter current water trap.
- Should have embedded memory storage for full raw data and facility for longer storage.
- Should be supplied with standard spares and accessories
- Standard should be complying with USFDA/BIS/CE and CDSCO.
- Should have ramp settings for CPAP Pressure.
- Should have mask size check feature
- Should have CPAP Pressure (4-20) cm of H₂O
- Should have display of at least 2 inches
- One-year warranty

15. BUBBLE CPAP

- Should be suitable for treating newborns with respiratory distress weighing 500-5000g.
- The CPAP pressure with oscillations should be generated by creating resistance in water column and bubbling of exhaled gas in the water column.

Should essentially have the following components,

A. Servo Heated Humidifier

- It should have servo controlled heated humidifier with following features,
 - i. Temperature and flow sensor with feedback mechanism.
 - ii. Monitoring temperature of gas at chamber end and near patient end.
 - iii. Display for temperature of saturated gas.
 - iv. Should have separate modes for invasive and non-invasive ventilation.
 - v. Should have alarms for high temperature /low temperature, low water in chamber, Heater adapter faulty/disconnect, Temp cum probe faulty/disconnect and hardware faults, The Humidifier should provide a visual indication as to the location of the fault.

B. Patient Delivery system

- i. The patient heating circuit should have integrated spiral heated coil for uniform heating.
- ii. Humidification chamber should have facility for auto feed and have a chamber compressible volume of at least 260-300 ml. The chamber should have a max peak flow of at least 180l/m.
- iii. Should be supplied with Nasal prongs/masks of silicon useful for babies weighing between 750-2500g. It should be provided as standard accessory. The nasal masks should be suitable for preterm and term babies. The mask should be soft and anatomically shaped.
- iv. Should be supplied with flexible nasal tubing which should be adaptable with the infants movement.
- v. The nasal tubing should enable prong or mask re positioning and keep straps away from the patients face.
- vi. Should be supplied with infant caps for 17-36cm circumference.
- vii. Should be supplied with nasal cannula which should be kink proof and have adhesive to secure on skin. The adhesive used should be neonate skin friendly.

C. CPAP Bubble Generator

- i. CPAP Bubble generator should have adjustable probe for pressure settings 3-10cm of H₂O. It should have detachable overflow container to maintain constant water level with a minimum volume of 500ml.
- ii. The system should have safety mechanism with pressure relief valve and ports for pressure and fio2 monitoring.

D. Air-Oxygen Blender

- I. Should have portable and lightweight and weight under 2 kg.
- II. Should have provision for adjusting fio2 from 21%-100%.
- III. The accuracy of the set value should be within $\pm 3\%$.
- IV. Should provide a output flow rate between 0-30LPM.
- V. The blender should be able to support an input supply pressure from 2-5 bar.
- VI. Should have inlets for connection of air and oxygen. Necessary tubing's and adapters to connect from the external gas supply sources (air and oxygen) may be supplied as standard accessories.
- VII. Should also have provision for connecting to central air and central oxygen outlets. Necessary tubing's and adapters/key plugs may be supplied in this regard.
- VIII. Necessary adapters towards mounting the blender on IV pole may be supplied as standard.
- IX. Should be supplied with a flow meter (0-15LPM) including the wing nut towards controlling the output flow.

E. ACCESSORY

- CPAP generator-1 No

F. CONSUMABLES

- i. Patient circuit with humidifier chamber – 20 Nos.
- ii. Nasal prongs (small, medium, large)-20each.
- iii. Nasal Masks (small, medium, large)-20 each.
- iv. Nasal Tubing-20 Nos.
- v. Nasal cannula-20 Nos.

- vi. Infant caps (small, medium, large)-20 Nos.
- vii. The cost of the above mentioned accessory and consumables' may be separately quoted which would be valid throughout the period of warranty.
- viii. Should be supplied with a trolley mounted on quality castors to fix the humidifier, CPAP generator, IV hooks, mounting brackets for gas supply lines etc. The required adapters/fixtures for mounting the same should be supplied as standard.
- ix. One-year warranty.
- x. Standard should be comply with USFDA/CE/BIS and CDSCO.

16. RADIANT WARMER

Sr.No.	Technical Specifications
	Performance/Electrical Specification
1	Should be microprocessor based temperature controller with Pre-warm, Manual and Skin Servo modes
2	Should have LED Display of Skin Temp., Air Temp., Set Temp. (Resolution 0.1°C) & Heater Power %.
3	Should have LCD to display modes of operation, Alarm messages, Skin Temperature in 'F and Timer.
4	Should be Thermistor based sensor with Accuracy of +/- 0.2°C (Temp. Calibration not require)
5	Should have following Modes of Operation: Skin Servo, Air, Manual and Pre-Warm
6	Should have Keypad lock facility
7	Skin mode Temperature range should be 34 to 38°C and resolution should be + 0.1 °C
8	Air mode Temperature range should be Room temp. to 39°C & resolution should be t 8 0.1°C
9	Manual mode should have Adjustable Heater Power 0 to 100% (with an increment of 9 5%) & Time duration:15mins
10	Pre-warm mode: Used to pre-warm the cradle before baby is placed on the bed. (No 10 alarm will be activated during Pre- warm mode) should have Adjustable Heater Power 0 to 25% (with an increment of 5%) & Time duration:30mins
11	Overhead FR-grade heater box should consist of Swivelling heater source for X- 11 Ray purpose & Consists of Ceramic heater (650 Watts) placed in parabolic reactor with a safety grill, With LED observation light for observing the baby
12	Should have Operating Voltage: 220-240 VAC, 50Hz
13	Power consumption should be around 800W (Max.)
14	Should consist of following Alarm: Audio alarm with 10 minute mute facility & 14 Skin OVER /UNDER RANGE, Air Hi/Low, Air Over temp. Skin Sensor failure, Air should have Visual alarms with message on LCD : Skin HI/LOW, Skin over temp, Sensor Failure, Heater failure, Power control failure.
15	Power Fail Alarm: Audio with LED indicator
16	Mechanical Specifications:
i	Should be Fixed height warmer with attached baby cradle mounted on 4" wheel

ii	Should have Stainless steel baby tray with water proof, zipped cover mattress of ii density approx. 21-25 kg/m ³ with Continuous head up/down facility
iii	Should have X-Ray cassette holder
iv	Should have Collapsible/lockable Side Supports made up of Unbreakable polycarbonate material
v	Should have Slots for sensor and ventilator tubing
vi	Should have Side tray for placement of monitor
vii	Should have Stainless steel infusion rod with facility to mount Syringe/infusion pump
viii	Should have Lower shelf below baby tray to facilitate under surface phototherapy
17	Should consist of Inbuilt T-Piece Resuscitator which should be manually operated, gas powered resuscitator designed to provide breaths at a set flow which ensures adequate pressure delivery and oxygenation to the infants. T-Piece device should have controlled PIP, consistent PEEP, thus providing the maximum oxygenation to the baby's lungs.
18	Specifications for T-Piece Resuscitation:
i	Gauge Range should be between -20 to 50 cm H ₂ O
ii	Manometer Accuracy should be ± 2 L/min
iii	Peak Inspiratory Pressure (PIP) @ 05L/min: 02 to 24 cm H ₂ O
	@ 08L/min :04 to 40 cm H ₂ O
	@10L/min:06 to 48 cm H ₂ O
iv	Peak End Expiratory Pressure (PEEP) @ 05L/min : 02 to 08 cm H ₂ O
	@ 08L/min:04 to 14 cm H ₂ O
	@10L/min : 06 to 20 cm H ₂ O
v	Gas Inlet Flow Range should range from 05L/min (min) to 10L/min (max)
vi	Maximum Pressure Release should be up to 50 cm H ₂ O
19	Dimensions should be as follows:
i	Height: 74.5" (1890 mm), Width : 28.0" (710 mm), Length : 39.0" (990 mm) Bed to Floor Height: 37.5" (950 mm)
ii	Distance between baby bed and heater: 33" (840 mm) Mattress size: 30" (762 mm) x 20" (508 mm)
iii	Height of side panel above baby bed: 5.75" (145 mm)
iv	1 year warranty
V	Standard should be complying with USFDA/BIS/CE and CDSCO

17. PHOTOTHERAPY MACHINE

- Should have heavy duty mobile stand phototherapy unit
- The light source should have height adjustment approx. 0.95 m to 1.5 m
- Light source should tilt - continuous up to +/- 90°
- Light source should have a cooling fan.
- Should consist of therapeutic high wattage blue LEDs, 16 nos. (with dimming facility)

- Wavelength should range from 455 to 465nm
- High irradiance at skin level should be 60 $\mu\text{W}/\text{cm}^2/\text{nm}$
- Should have effective surface area 50 cm x 30 cm at 45/35 cm from light source with uniformity ratio of 0.40
- LED Lamps should be rated to last upto 50,000 hrs (with 30% depreciation)
- Should consist of high wattage white LEDs, 4 nos., for observation purpose.
- Should have dual digital cumulative hour timer for LED usage and patient exposure
- The system should have overheat protection power cutoff for temperature at 90°C
- The unit should be CE/ BIS/ USFDA certified
- 1 year warranty

18. OXYGEN CONCENTRATOR 10LTR-DUAL FLOW

- Oxygen flow at 10 lts per minute
- Dual flow.
- Oxygen concentration 93%+3%
- Oxygen output pressure 0.04-0.07 Mpa
- With oxygen number display
- LED lights Display
- Safety alarms for-low oxygen, power off, compressor fault, overheating.
- 1 year warranty
- Should be complying with USFDA/BIS/CE and CDSCO and ISO

19. FULLY-AUTO BIOCHEMISTRY ANALYZER (200 THROUGHPUT)

Throughput	Double Reagent : 200 Tests/hour
Analysis Method	End Point, Fixed Time (2 point), Kinetic Rate A, Kinetic Rate B
Assay modes	Colorimetric, Turbidimetry, Single & Double Reagent, Multi standard, Mono & Bi-chromatic
Sample Volume	2 -35 μl , 0.10 μl step
Sample Dilution	Dilution ratio of 2 to 40
Reagent Position	30 onboard cooled reagent positions maintained by water cooled peltier system
Reagent Volume	100-360 μl
Reaction Cuvette	cuvette 57
Cuvette Washing	On board laundry with 7 step washing system
Water Consumption	2.5 -4 Litres/hour
Optical length of cuvette	5mm
Reaction Volume	180-550 μl
Reaction Time	300 to 600 sec (Depending on the designated cycle time and number of reagents)
Reaction Temperature	37 $\pm$ 0.2°C
Measurement	Photometry

Light Source	Halogen Tunsten lamp
STAT	User defined STAT function
Wavelength	9 interference filters: 340, 405, 450, 505, 510, 546, 578, 600, 620, 660, 700 nm with Fibre optics (Must have Fibre optics to ensure maximum intensity of light reaches to the detector and also increases the life of filter)
Absorbance Range	0 -3.0 Abs
Resolution	0.0001 Abs
Relex	User defined Reflex testing
STAT	User defined STAT function
Calibration	K-Factor, Linear (one, two and multi point), Logit-log, Spline, Exponential, Polynomial (second, third and fourth order)
Calibration Points	Multipoint curves for up to 6 points
Auto re-run	Execution by repeat run list or auto execution
	Auto execution according to abnormal marking or range over
	Reduced/increased volume repeat run also possible
Data Storage	Test results: 1,000,000
	Reaction curve: 40,000 tests
	Profiles: 20 per sample type
Quality Control	Within day as well as day-to-day X and X-R control diagram
	(L-J Graphs)
	Real time quality control based on Multi-rule method
	Mean, SD, %CV, R is calculated for all parameters for sample replicates
Power Supply	AC 220 V $\pm$ 10%, 50 Hz or AC 110 V $\pm$ 10%, 60 Hz
Temperature	10°C -30°C
Humidity	40 - 80% free from water dew formation
Operation System	Windows XP or Windows 2007 or Winfix
Input	RS 232 interface/computer
Output	Multi Format Printout
Dimensions	550 mm (W)x 420 mm (D) x 380 mm (H)
Weight	Approx 25 kgs
Certificates	CE,ISO 13485
Available Certification & Conditions	Reagents should be photometric, parameters like D- Dimer, Ferritin procalcitonin and other parameter for covid.
	All reagents should be manufactured in India and must have its own manufacturing units for reagents in India

	The company supplying the Instrument must have more than 100 units installation base in India
Warranty	One year.

20. CTG MACHINE WITH DUAL PROBE

- All items must be from one manufacturer for smooth compatibility and working of the system. All the equipment's should have CE/USFDA/BIS and CDSCO from same manufacturer
- It should be Android based wireless CTG with twin monitoring
- It should-have no antennas so that wireless CTG unit can be kept in user-and patient can sit for monitoring having all probes in other adjacent room means concrete wall should not stop the monitoring
- It should have screen size of more than 10 inches with High resolution Hi-tech color TFT Screen with tilt adjustment up to 90 degree so that the monitor can be seen from any angle by user or the patient clearly
- It should have Wireless probes (No wires with FHR, TOCO & Movement Marker)
- The FHR probe should be water proof, rechargeable, should show the battery life in FHR probe, battery life should be 8-10 hours for non-interrupted monitoring as it support longtime work& also should
- show the FHR reading in probe& FHR range should be 30-240 bpm
- It should be Touch Screen functions, easy to operate, Long Life& user friendly
- It should be CTG Scoring Facility
- It should be Automatic CTG Reporting with scoring by FISHER, KREBS, ACOG, NS-CTS
- It should be High sensitive transducer for FHR detection with 14 Elements / Crystals or more
- It should be Broad beam technology & It should have 1-3MHz Pulse Doppler transducer
- The wireless CTG unit should have minimum Battery Backup of 4 hours
- It should have Inbuilt 152 size Thermal Printer, print facility to be print on A4 size paper, data should transfer in USB & Patient storage data with multiple patients ID of more than 30000 hours with playback & printing facility Basic Parameters: FHR, TOCO, Event Marker
- It should be Interface of Parameters of FHR, TOCO, FM with waveform& digital display simultaneously
- It should be Real time, accurate & reliable result for better diagnosis & monitoring
- It should be Compact design, extremely light weight, less than 1.8 Kg so that it is easy to move & taking less space
- It should have alarm for Tachycardia, Bradycardia, lack of paper, low battery power, network disconnection, probe off & high low limit alarms
- It should have Central Nursing Station facility to connect with other bed side monitor by Bluetooth or WiFi.
- One year warranty.